Marking Memo: Euler Method Task

1. Understanding the Euler Method

- Student shows derivation from derivative to discrete Euler form (10 pts)

2. Excel Implementation

- Well-labeled columns and logical layout (5 pts)
- Correct update formula using Excel syntax (5 pts)
- Correct use of relative cell references (5 pts)
- Correct and labeled graph (2 pts)
- Explanation of how each column functions (3 pts)

3. Python Implementation

- Code matches update rule; no logic errors (5 pts)
- Correct use of matplotlib with labels (2 pts)
- Correct initialization and update of arrays (2 pts)
- Readable code with comments (2 pts)
- Correct stopping condition (4 pts)

4. Insight and Reflection

- Comparison of Excel vs Python (5 pts)
- Why numerical methods matter (5 pts)

5. Lab Report Key Results to Investigate

- What time does it take for the temperature to reach approximately 0 degC when lambda = 1? (3 pts)
- What trend do you observe in the cooling rate as lambda changes from 0.1 to 5? (3 pts)
- What is the difference (if any) between the analytical and numerical results for m = 1 kg? (3 pts)
- How does the mass of the object affect the time it takes to reach 50 m? (3 pts)
- Can you derive and explain the analytical solution x(t) = x0 + v0*t + 0.5*(F/m)*t^2? (3 pts)

6. Written Application of Euler Method

- Describe a real-world physical system not covered in class that can be modeled with an ODE (4 pts)
- Explain how Eulers method would be applied to solve this ODE (3 pts)

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- Discuss limitations or assumptions of using Eulers method in this case (3 pts)

Total: 100 points